Cellular Automata for Two Lane Traffic Flow

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Abstract

Key Words:

Contents

1 Introduction

1.1 Real-world Context

Urban Traffic Congestion is a pervasive issues across a range of different demographics, whether it is in a developing or developed country. Traffic Congestion has always been a significant concern in traffic control mechanisms caused by an unbalanced demand-supply situation that directly affects the problem of large fuel consumption and greenhouse emission. Congestion leads can lead to a range of issues such as traffic jams, unhealthy air quality, vibrations, noises and accidents which are significant problems for the environment and human health [1].

In the UK, this issue is particularly prevalent. In the 2019, UK drivers lost an average of 115 hours each to road traffic congestion [2].

To reduce congestion, we should look at optimising traffic flow, this can be done through the use of traffic lights, by employing various strategies we can maximise the flow of traffic and reduce congestion. Since traffic lights are already widely used, it is important to understand how they can be optimised to improve traffic flow and reduce congestion allowing for minimal costs.

1.2 Why discrete cellular automata

We can leverage the use of cellular automata to model traffic flow, this is because they are a powerful tool for simulating complex systems and can capture the dynamics of traffic flow in a simple and efficient way. Cellular Automata also allow us to assess the impact of different traffic light strategies on traffic flow, and can help us to identify the most effective strategies for reducing congestion.

Traffic cellular automata models have been widely used in traffic flow research, and have been shown to be effective in capturing the dynamics of traffic flow and congestion. Examples of cellular automata models include the Nagel-Schreckenberg (NaSch) model and the Biham-Middleton-Levine (BML) model [3, 4].

For this study, we will use the NaSch model due to its simplicity and more clearly explaining the effects on traffic lights in the system. With this model we can easily consider the effects of multiple traffic lights on the system as well as the effects of different traffic light strategies on traffic flow and congestion. Additionally we can also consider how a second lane can affect the traffic flow and congestion in the system with different vehicle types taken into consideration.

1.3 Aims and Research Questions

In this report, we will explore three different contexts of traffic flow, exploring how different traffic light layouts effect the system. The first context will be a single lane system, where we vary traffic light number and traffic light strategies [3]. The second one will be a two lane system, where will explore the effect of varying traffic light number and traffic light strategies with the introduction of different vehicle types on the system [5, 6]. The last one is a perpendicular intersection in which two single lane roads intersect with each other, we will explore the effect of varying traffic light strategies on the system [7]. The unifying question across these two is light configuration and strategy, we want to understand how different traffic light configurations and strategies can affect traffic flow and congestion in both single lane and two lane systems. The three geometries are shown in Figure 1.

We will address four research questions:

1. How does single-lane through-flow scale with the number of traffic lights, the cycle length, the green fraction and the inter-light phase offset?
2. When does green-wave coordinate outperform synchronised lights and why?
3. Does symmetric lane changing complement the green-wave or fight it?
4. How do the above results change when we consider a perpendicular intersection?

2 Background

2.1 Continuum vs Discrete Traffic Flow

Analytical traffic models can be either continuum or discrete. Early examples of traffic flow models were based on continuum mechanics Continuum models treat traffic as a continuous fluid, where instead of modelling individual vehicles, the vehicle density, mean velocity and flow are all modelled as continuous functions of space and time. Lighthill and Whitham represented the road as a one-dimensional fluid with flow q (vehicles per hour), concentration k (vehicles per mile) and posited as the fundamental hypothesis of their theory that the flow q is a function of k alone [8]. The continuum models are useful for understanding traffic flow at a macroscopic scale, formation of traffic jams and the effects of traffic lights but cannot capture the microscopic behaviour of individual vehicles and the interactions between them. Discrete models on the other hand address this weakness by treating everything as a finite integer such as space, time, velocity and the vehicles themselves, as we discuss next.

2.2 The NaSch cellular automaton

Nagel and Schreckenberg proposed a stochastic discrete cellular automaton where a road is represented as a one-dimensional lattice of cells, each of which can be occupied by a single vehicle or be empty. Each vehicle carries an integer velocity v in the range $\{0, 1, \ldots, v_{\max}\}$ and the system is updated in discrete time steps according to a set of rules that govern the movement of the vehicles. The rules are as follows:

1. **Acceleration**: If the velocity of a vehicle is less than the maximum velocity $v_{\max}$, and if the distance to the next car is larger than $v + 1$, the velocity is increased by 1, $[v \to v + 1]$.
2. **Deceleration**: If a vehicle at cell i sees the next vehicle at cell $i + j$ (with $j \leq v$), it reduces its speed to $j - 1$, i.e. $[v \to j - 1]$.
3. **Randomisation**: With probability p, the velocity of each vehicle (if greater than zero) is decreased by one $[v \to v - 1]$.
4. **Movement**: Each vehicle is advanced v cells.

Here p is the probability of random deceleration, which is a key parameter that controls the behaviour of the system. Through the four rules above, the general properties of single lane traffic are modelled on the basis of integer valued probabilistic cellular automaton rules. This simple model is effective in capturing the nontrivial and realistic behaviour of traffic flow. Step 3 is particularly important for simulating realistic traffic flow, without it the dynamics is purely deterministic, and from any initial configuration the system relaxes to a stationary pattern shifted backwards one cell per timestep [3, 9]. The model also accounts for natural velocity fluctuations due to human behaviour or due to varying external conditions. The four steps above are unchanged across the single lane, two lane and intersection scenarios.

2.3 Two-Lane Extensions

Rickert et al. proposed a two-lane extension to the NaSch model, where vehicles can change lanes according to a set of rules that depend on the local traffic conditions. The lane-change sub-step before the NaSch update with both sub-steps running at the start of each time step. The lane-change rules are as follows [5]. Chowdhury et al. further extended this to include two different types of vehicles, slow (e.g. a truck) and fast (e.g. a car) [6].

2.4 Intersection Extension

2.5 Optimising Signalised Systems

3 Model Description and Boundary Form

3.1 Model Assumptions

3.2 Model Scaling and Interpretation

The model can be scaled so that each car occupies around 7.5 m of road which is the length of one cell. As the average velocity of cars in free traffic is 4.5 cells per time step it should correspond to around 120 km/h (in Germany), which is a reasonable speed for free traffic on a motorway, naturally we arrive at a time for iteration of

$$7.5\frac{\text{m}}{\text{cell}} \times 4.5\frac{\text{cells}}{\text{time-step}}/(120/3.6)\frac{\text{s}}{\text{m}} \approx 1.0 \text{ s} \tag{1}$$

[3]. As this model is scalable to different scenarios there is not a need to non-dimensionalise the model, we can simply use the above scaling to interpret the results in real world terms

3.3 Road Geometries and Light Configuration

The three road geometries studied in this report are shown in Figure 1.

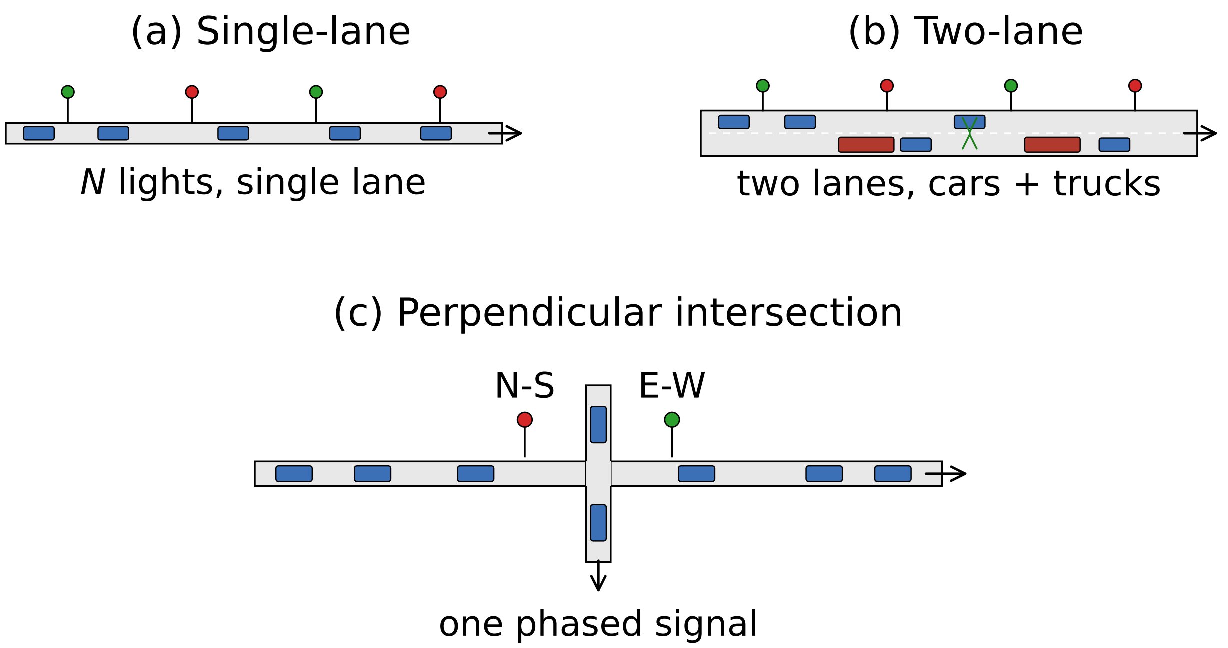

Figure 1: The three road geometries studied in this report. (a) Single-lane road with N traffic lights, cars travel left-to-right. (b) Two-lane road with mixed vehicle types (cars in blue, trucks in red) and s N synchronised lights, the green arrow marks a candidate symmetric lane change. (c) Perpendicular crossing of two single-lane roads with one phased signal at the junction, lights are shown in their current phase (green for the east-west direction, red for north-south). Cell counts, light spacing, and cycle parameters are set in the corresponding model section. Schematics are illustrative

3.4 Single Lane Lattice and State

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